Stablecoins – Depeg Prediction Analysis

Machine Learning-Based Early Warning Systems Design

Rob Springett, Jill Wang, Ara Ludwig, Ify Enwezor

Carnegie Mellon University

MSBA Capstone – 46899-M

Professor Ganesh Mani

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Abstract

Stablecoins have emerged as a critical infrastructure layer of the cryptocurrency ecosystem, functioning as a reliable medium of exchange and store of value for trillions of dollars in daily transactions. Despite their design intent, stablecoins are not immune to losing their $1.00 peg — events that can trigger cascading losses across traders, protocols, and the broader market, while causing long-term damage to issuer credibility and user trust. The ability to anticipate these depeg events before they manifest in open markets represents a significant opportunity for risk management and market stability.

This paper presents a machine learning-based early-warning system capable of predicting stablecoin depeg events hours in advance. We constructed a dataset spanning seven stablecoins from 2017 to early 2026, drawing on four broad source categories: stablecoin price feeds, on-chain mint and burn activity, DeFi liquidity pool flows, and macro market indicators — all sampled or resampled to a common 5-minute resolution. Using modern feature engineering and model selection techniques, we trained a predictive model on over 3.3 million observations, including data from historical failures such as TerraUSD, enabling the system to identify early stress signals in on-chain flows and market dynamics that precede a peg breakdown. Model performance was validated using a strict walk-forward time split and a Leave-One-Event-Out robustness test across known depeg events.

The result is a practical risk tool that transforms stablecoin monitoring from a reactive process into a forward-looking signal, with direct applications for stablecoin issuers, risk managers, and protocol operators.

Acknowledgements

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Introduction

Stablecoins occupy a unique and increasingly consequential position within the digital asset ecosystem. Designed to maintain a stable value relative to a reference asset—most commonly the US dollar—they have rapidly become the connective tissue of decentralised finance (DeFi) and a critical instrument in cross-border payments. Yet their apparent stability masks profound structural vulnerabilities.

Research Question

This paper examines the following central question:

> *Can stablecoin depeg events be predicted using machine learning models, and if so, which features and model architectures provide the greatest predictive accuracy?*

This primary question is further decomposed into four sub-questions that guide the structure of the analysis:

1. What is the key on-chain and off-chain indicators that precede a stablecoin depeg event historically?
2. Which machine learning approaches are best suited to early warning classification in this context?

Significance of the Study

Stablecoin issuers face a unique challenge: their core product is trust, and that trust can unravel within hours. A depeg event, even at a shorter timeframe, can trigger redemption waves, like a run on the bank in traditional finance, and can impact market share for the longer term.

For fiat-backed issuers including Ripple, Tether and Circle the ability to predict emerging stress gives time to act before the open market reacts; pre-positioning liquidity, engaging market makers, and managing reserve drawdowns in an orderly way rather than under pressure. At its core, predicting a depeg is not just about avoiding a bad outcome; it is about protecting the fundamental promise that one token is always worth one dollar.

Stablecoin Structures Summary

Stablecoins can be broadly categorized into three structural types: fiat-collateralized, crypto-collateralized, and algorithmic. The seven stablecoins examined in this study span all three categories and illustrate the range of design approaches, reserve mechanisms, and risk profiles relevant to depeg prediction.

Tether (USDT)

USDT is the largest stablecoin by market capitalization, maintaining a 1:1 dollar peg through reserves held by Tether Limited. Despite persistent questions over reserve composition and transparency, it has retained its peg through multiple periods of market stress, making it a key benchmark in this study.

USD Coin (USDC)

Issued by Circle, USDC is widely considered one of the more transparent fiat-backed stablecoins, supported by monthly third-party attestations. Its brief depeg to $0.87 in March 2023 — triggered by $3.3 billion in reserves held at the collapsed Silicon Valley Bank — demonstrates that custodial and counterparty risk can affect even well-collateralized instruments.

Ripple USD (RLUSD)

Launched in late 2024, RLUSD is Ripple's regulated enterprise stablecoin, backed by US dollar deposits and short-duration government securities, available on both the XRP Ledger and Ethereum. Designed for institutional cross-border payments, it represents a newer generation of compliance-first stablecoins with a short but auditable operating history.

Binance USD (BUSD)

BUSD was a fiat-backed stablecoin issued by Paxos in partnership with Binance, wound down in 2023 following NYDFS regulatory action and an SEC Wells Notice. Its discontinuation highlights that stablecoin failure can be regulatory rather than market-driven — a risk dimension distinct from collateral or liquidity stress.

Ethena USDe (USDe)

USDe is a synthetic dollar that maintains its peg via a delta-neutral strategy, pairing long spot ETH positions with short ETH perpetual futures rather than fiat reserves. While yield-generating, this structure is sensitive to negative funding rates and exchange counterparty risk, warranting distinct feature engineering in predictive models.

TerraUSD (UST)

UST was an algorithmic stablecoin pegged through a mint-and-burn relationship with the LUNA token. Its collapse in May 2022 — which erased over $40 billion in value within days — remains the definitive case study in algorithmic stablecoin failure and provides the most significant labelled depeg event for model training in this study.

Dai (DAI)

DAI is a decentralized, crypto-collateralized stablecoin issued by MakerDAO, generated by locking ETH and other assets in over-collateralized smart contract vaults. Its governance-controlled stability fees and dynamic collateral requirements have supported peg resilience across multiple market downturns, though its stability remains exposed to collateral volatility.

Stablecoin On-Chain Transactions

Curve Finance Liquidity Pools

Curve Finance is a decentralized exchange optimized for low-slippage trading between stablecoins and similarly priced assets. Unlike general-purpose AMMs, Curve uses a specialized invariant formula that concentrates liquidity around the peg price, making it the dominant venue for stablecoin-to-stablecoin swaps and a critical infrastructure layer for peg maintenance. Its flagship 3pool — comprising USDT, USDC, and DAI — serves as a primary benchmark for stablecoin liquidity health across the DeFi ecosystem.

Stablecoin Minting and Burning Mechanisms

The processes by which stablecoins are created (minted) and destroyed (burned) are fundamental to understanding how their pegs are maintained and where vulnerabilities arise. Minting and burning mechanisms differ significantly across stablecoin types and directly influence the speed and reliability of peg restoration following a deviation.

Fiat-Collateralized (USDT, USDC, RLUSD, BUSD)

In fiat-backed models, minting occurs when an authorized user or institution deposits US dollars with the issuer, who in turn issues an equivalent quantity of tokens on-chain. Burning is the reverse: the holder redeems tokens with the issuer, who destroys them and returns the corresponding fiat. This bilateral process is typically restricted to institutional counterparties, meaning retail users must rely on secondary market trading to enter or exit positions. Arbitrageurs play a critical role in peg maintenance — when a stablecoin trades below $1.00, they purchase tokens on the open market and redeem them at par with the issuer, profiting from the spread and restoring the peg. The speed and reliability of this arbitrage loop is therefore a key determinant of peg resilience and a meaningful input feature for predictive models.

Algorithmic (UST)

Algorithmic stablecoins like UST relied on open-market mint-and-burn arbitrage rather than direct fiat redemption. Any holder could burn $1.00 worth of LUNA to mint 1 UST, or burn 1 UST to receive $1.00 worth of newly minted LUNA. In theory, this guaranteed a floor price of $1.00 through arbitrage. In practice, the mechanism created a reflexive feedback loop: as UST depegged, the volume of UST burned for LUNA flooded the market with new LUNA supply, collapsing its price, which in turn eroded the $1.00 redemption value and accelerated further UST selling. This death spiral dynamic is a structural property of uncollateralized algorithmic designs that no arbitrage mechanism could arrest once confidence was lost.

Crypto-Collateralized (DAI)

DAI is minted when a user locks approved collateral — ETH, wrapped Bitcoin, or real-world assets — into a MakerDAO vault at an over-collateralization ratio, typically 150% or higher. To recover their collateral and burn the corresponding DAI, users repay the minted amount plus an accrued stability fee. If the collateral value falls below a liquidation threshold, the vault is automatically liquidated: the collateral is auctioned and the proceeds used to burn the outstanding DAI.

Synthetic (USDe)

USDe is minted when a user deposits approved collateral — primarily stablecoins or ETH — with Ethena Labs, which simultaneously opens an equivalent short position in ETH perpetual futures to create a delta-neutral portfolio. Redemption burns USDe and returns the collateral after unwinding the corresponding hedge.

Literature Review

A growing body of scholarship examines the stability properties and systemic risks of stablecoins. We surveyed online literature for definition and modeling for depeg event, the key findings are below.

Depeg Definitions: Threshold and Duration

The dominant academic standard is a ±1% deviation from $1.00, measured on daily closing prices (Lyons & Viswanath-Natraj 2023; Grobys et al. 2021; Gorton & Zhang 2021). For fiat-backed stablecoins, Bertsch (2023, BIS) and Anadu et al. (2023, NY Fed) argue ±0.5% is more appropriate — any sustained deviation beyond that signals redemption friction, not routine liquidity imbalance. For crypto-collateralized DAI, Nadler & Schär (2020) recommend ±2% given inherent collateral volatility.

A key refinement across multiple papers is a duration filter: Anadu et al. require at least 3 consecutive hours outside the threshold; Klages-Mundt & Minca (2022) formally distinguish transient (< 24h) from persistent (≥ 24h) depegs, arguing only the latter represent systemic risk. Another key insight is that using single-exchange prices overstate severity — during the USDC SVB crisis, Coinbase reported a low of $0.877 while the cross-exchange VWAP only fell to ~$0.972. Caramichael & Liao (2022, Federal Reserve) and Lyons & Viswanath-Natraj (2023) both recommend cross-exchange VWAP, which is what we use via CoinAPI's IDX_REFRATE_VWAP.

What We Do Differently

This project advances the literature on three dimensions:

1. Resolution at 5-minute rather than daily, enabling early-warning signals
2. On-chain leading indicators — mint/burn flows, treasury inflows/outflows, and Curve pool swap imbalances
3. Pooled cross-coin modeling — including failed (UST) and discontinued (BUSD) coins to maximize real depeg examples

Defining a Depeg Event

A precise and operationally consistent definition of a depeg event is a prerequisite for any machine learning approach to stablecoin stability prediction. Without a clearly bounded target variable, model training, evaluation, and comparison across studies becomes unreliable. Despite its centrality to the literature, there is no universally accepted definition, and existing research varies considerably in how depeg events are identified and classified.

Price Deviation Threshold

The most common approach defines a depeg event as a sustained deviation of a stablecoin's market price from its target peg — typically \$1.00 — beyond a specified threshold. Thresholds used in the literature range from 0.5% to 5%, depending on the stablecoin type and the study's focus. For the purposes of this paper, a depeg event is defined as any instance in which a stablecoin's price deviates from its \$1.00 target by more than 1% (i.e., trades below \$0.99 or above \$1.01) for a sustained period of at least one hour, as recorded by high-frequency price feed data. This threshold reflects the tolerance commonly observed in market-making activity and distinguishes genuine stress events from routine bid-ask spread noise.

Severity Classification

Depeg events can be further classified by severity to enable more granular model labelling. A minor depeg refers to a deviation of between 1% and 5%, typically self-correcting within hours through arbitrage. A moderate depeg involves a deviation of 5% to 20%, often requiring active intervention by the issuer or protocol governance. A severe depeg — such as that experienced by UST in May 2022 — represents a deviation exceeding 20%, with the potential for a self-reinforcing collapse. This three-tier classification is adopted as the labelling framework for the machine learning models developed in this study.

Duration and Persistence

Price deviation alone is insufficient to characterize a depeg event without consideration of duration. Flash depegs — brief deviations caused by low-liquidity periods or large single transactions — are qualitatively distinct from structural depegs driven by fundamental loss of confidence or collateral insufficiency. This study applies a minimum duration filter of one hour to exclude transient price dislocations and focuses predictive modelling on events with meaningful systemic implications.

Our Depeg Definition

For the purposes of this project, a depeg event is defined as a sustained deviation of a stablecoin's price from its $1.00 peg. Specifically, we require the absolute price deviation to exceed 0.5% — equivalent to a price below $0.995 or above $1.005 — for a minimum of three consecutive 5-minute bars, representing a 15-minute sustained breach. This persistence requirement is intentional: it filters out transient tick noise and brief liquidity-driven price dislocations that self-correct within a single bar, ensuring that only meaningful, sustained stress events are labeled as depegs.

Methodology

Our analytical pipeline follows a centralized-then-divergent architecture. Data collection, descriptive analysis, and feature engineering are performed once on the full seven-coin panel, producing a shared foundation from which both modeling efforts draw.

From this common base, the pipeline splits into two independent pods, one targeting depeg events in either direction, the other focused exclusively on below-peg breaks with each conducting its own feature selection, model selection, and hyperparameter tuning against its specific prediction objective. This design avoids duplicating the substantial data infrastructure while allowing each pod to optimize independently for the characteristics of its target

Data Sources

The dataset integrates eight sources across four categories — peg price, on-chain supply and flows, DEX trading activity, and macro context — unified at five-minute resolution. It covers seven stablecoins (USDT, USDC, DAI, BUSD, UST, USDe, and RLUSD) from their respective launch dates through February 28, 2026. Daily sources are forward filled into five-minute bars.

Peg Price: CoinAPI

Five-minute OHLCV data is sourced from CoinAPI's IDX_REFRATE_VWAP index, a volume-weighted average price aggregated across all major exchanges.
https://docs.coinapi.io/indexes-api/index-offerings/vwap-index

On-Chain Supply and Treasury Flows

1. Ethereum mint and burn (Etherscan). ERC-20 Transfer events to and from the zero address are collected for USDT, USDC, DAI, BUSD, USDe, and RLUSD via the Etherscan V2 API, aggregated to five-minute mint count, burn count, and net flow.
2. USDT treasury flows — Ethereum and TRON (Etherscan, TronGrid). When institutions redeem USDT at par they send tokens directly to Tether's treasury wallet rather than selling on the open market. Inflows to the known Ethereum treasury address and three TRON treasury wallets are tracked separately.

3. USDC Solana (Dune Analytics, Helius). Circle issues USDC natively on Solana. Historical mint and burn data is collected via Dune Analytics (tokens_solana.transfers), with incremental updates via the Helius API, deduplicated on transaction hash.
4. RLUSD XRPL (Dune Analytics). Ripple issues RLUSD natively on the XRP Ledger. Mint and burn payments are collected via Dune's XRPL dataset.
5. USDT Omni Layer**.** Prior to ERC-20, USDT operated on the Bitcoin Omni protocol.

DEX Trading Activity: Curve Finance

1. Token Exchange events are collected from five Curve pools via the Etherscan V2 API, providing sold volume, bought volume, and net sell volume per token per five-minute bar.

Market and Macro Context

BTC/USDT and ETH/USDT five-minute OHLCV data is sourced from the Binance public API from August 2017. For USDC, BTC/USDC, ETH/USDC, and USDC/USDT pairs are also collected as relative stress indicators.

FRED and Alternative.me. Four daily macro series are collected via the FRED API: the US Dollar Index (DXY), the CBOE Volatility Index (VIX), the 10-year US Treasury yield, and the Federal Funds rate. The Crypto Fear and Greed Index is sourced from Alternative.me. All are forward-filled to five-minute resolution, providing the macro backdrop against which stablecoin stress is interpreted.

Recommended Data Sourcing Improvements

One data point we were unable to source was Centralized Exchange orderbook and trading volumes. These datasets are available from many vendors (COINAPI, Coindesk) but are only accessible with subscriptions with costs that the budget of our study.

Descriptive Analysis

Per Coin Analysis

USDT — Tether (Fiat-backed, 2017–2026)

Tether is the oldest and most liquid stablecoin in the dataset, with 897,936 five-minute bars spanning August 2017 to February 2026. Its 71-column feature set includes Binance CEX microstructure data, though the USDC/USDT pair on Binance did not exist before December 2018 and a 166-day structural gap exists between September 2022 and March 2023. These windows are preserved as NaN rather than filled, allowing CatBoost to handle them natively without introducing false price signals. The Fear and Greed index is filled with 50 — the neutral midpoint — for the pre-February 2018 period where it did not yet exist.

Under normal market conditions USDT holds within plus or minus 0.1% of its dollar peg, with more than 99% of all bars falling within plus or minus 0.3%. The distribution is strongly leptokurtic — an extremely narrow central peak flanked by heavy tails that correspond to crisis periods. The directional pattern is clear: USDT episodes overwhelmingly depeg below the dollar, reflecting redemption pressure that overwhelms arbitrage during stress. Three distinct stress windows are visible in the 30-day rolling depeg rate: the October 2018 reserve scare, the March 2020 Black Thursday crash, and the May 2022 UST contagion period.

The October 2018 episode remains USDT's worst on record, with the price reaching a minimum of approximately $0.921, an 8% deviation sustained for around 48 hours. Burn volumes spiked simultaneously and the Binance spread widened to three to four times its normal level, consistent with market makers reducing inventory in response to genuine reserve uncertainty. The March 2020 episode was shallower at roughly 3% maximum deviation, driven by the COVID-induced ETH crash rather than reserve concerns, and resolved more quickly as arbitrageurs stepped in once collateral values stabilized. The May 2022 contagion from the UST collapse was more sustained, with around a 4.8% maximum deviation lasting approximately 72 hours and peer depeg rates across USDC and DAI spiking simultaneously.

Severity episodes are predominantly Mild for USDT — brief, self-correcting excursions — with Moderate clusters confined to the three crisis windows and no standalone Severe episodes across the full nine-year history. In terms of temporal structure, there is a mild overnight elevation between 00:00 and 06:00 UTC consistent with thinner liquidity, a slight weekend premium, and a clear year-level pattern with 2022 representing the peak risk year by a substantial margin.

The strongest univariate predictive signals for USDT are rolling price deviation statistics, particularly the one-hour rolling standard deviation with an adjusted AUC of 0.82 to 0.90, confirming that volatility becomes elevated before the formal threshold is crossed. The count of bars above 0.3% in the prior hour adds complementary information by capturing sustained soft crossings rather than single spikes. USDT also uniquely benefits from Binance CEX microstructure features: spread widening and shifts in the taker buy ratio precede price moves by enough to be actionable within the five-minute bar cadence. On-chain burn volumes on Ethereum round out the signal, with redemption outflows detectable on-chain before the price responds.

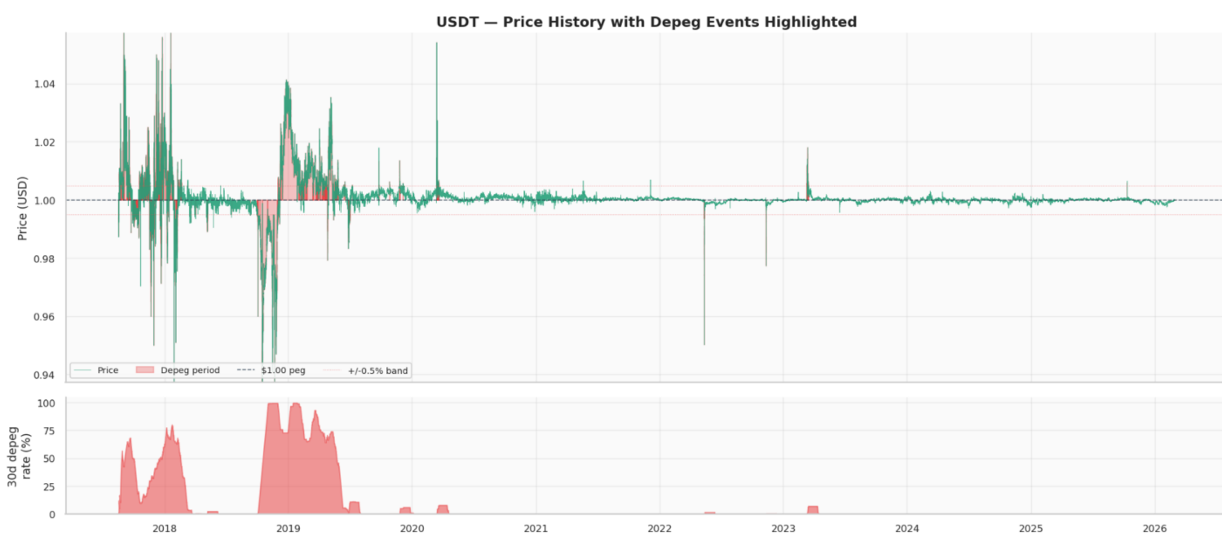

Figure 1. USDT Pricing History

USDC — USD Coin (Fiat-backed, 2018–2026)

USDC has 775,506 rows covering October 2018 to February 2026, with 88 feature columns reflecting its richer on-chain data availability. Its normal operating range is tighter than USDT at plus or minus 0.05 to 0.10%, and for most of its history the coin has been remarkably stable. However, a single event — the March 2023 SVB bank run — produced a depeg of approximately 13% that lasted around 80 hours and dominates the entire distributional picture. The result is a bimodal price deviation distribution: a sharp, narrow peak at zero representing normal operation and a pronounced left tail corresponding to the SVB event.

This bimodality reflects something fundamental about USDC's risk profile. Unlike USDT, where stress accumulated gradually through redemption dynamics, the SVB depeg was triggered by a discrete news event: Circle's disclosure that $3.3 billion of its reserves were held at Silicon Valley Bank on the day of FDIC seizure. The price remained essentially flat throughout March 10 despite substantial on-chain signals building, then dropped from approximately 0% deviation to below -2% within hours of the formal depeg onset and continued falling to -12% over the following 24 hours before recovering fully on the Federal Reserve backstop announcement.

The SVB case study is particularly valuable because it illustrates the lead-time advantage of on-chain flow features over price signals. The net flow z-score — a 30-day normalized measure of the mint/burn balance — crossed below the -3 sigma threshold approximately 21 hours before the formal depeg onset. This was not a one-off spike; the signal oscillated below -3 sigma throughout March 10, indicating sustained redemption pressure. In the 24 hours before the price break, mint and burn volumes showed a distinctive two-phase pattern: an initial large mint-and-burn cycle consistent with market makers cycling inventory in response to early stress signals, followed by accelerating burn volumes with minimal offsetting minting. Price deviation, by contrast, remained near zero throughout this entire period, confirming that on-chain flow features provide genuine leading information that is not available in price data alone. This validates the design decision to include on-chain flow features as primary model inputs and confirms the one-hour prediction horizon as a practical early warning window for this class of event.

The May 2022 UST contagion also affected USDC, producing a Moderate episode with a maximum deviation of approximately 2.5% lasting around 36 hours. In this case the fiat reserves were never in question — the stress reflected redemption queue pressure as investors exited stablecoins broadly — which explains why the episode was significantly shallower than SVB and resolved more quickly. The strongest predictive signals for USDC mirror USDT in structure, with rolling price volatility and bars above the 0.3% threshold leading the ranking, supplemented by Ethereum burn volumes, Curve 3pool volume z-score, and the Fear and Greed index.

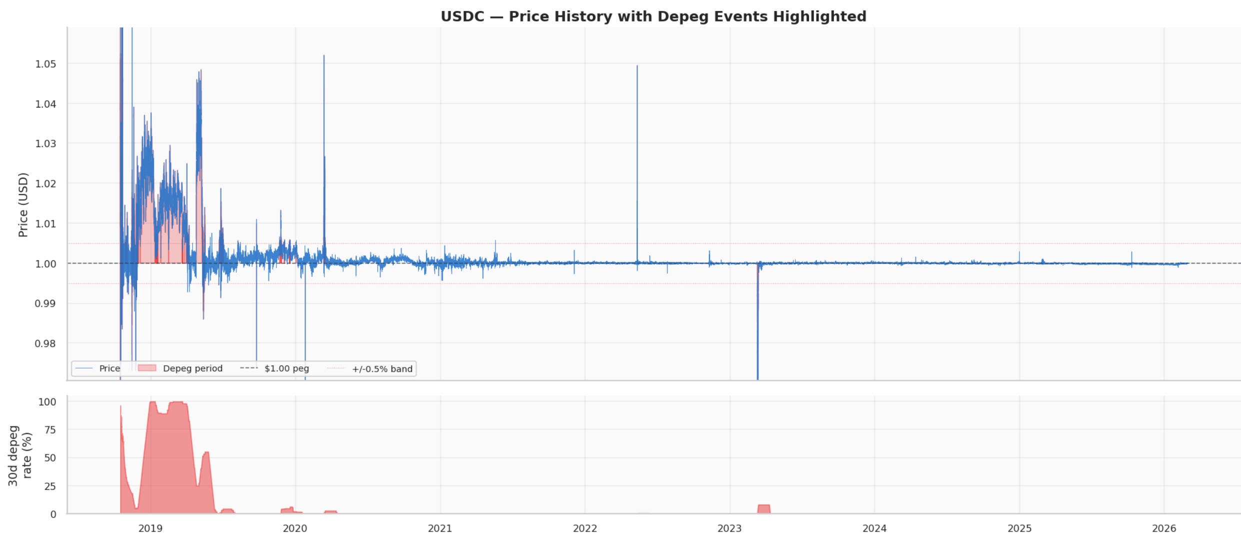

Figure 2. USDC Pricing History

DAI — MakerDAO (Crypto-backed, 2018–2026)

DAI has the highest sustained depeg rate among surviving coins at 3 to 5% of five-minute bars, reflecting a fundamentally different risk mechanism. With 828,963 rows covering April 2018 to February 2026, it spans three distinct structural regimes: single-collateral DAI backed solely by ETH before 2019, multi-collateral DAI from 2019 to 2021, and the PSM era from 2021 onward in which a Peg Stability Module enables 1:1 USDC swaps that have dramatically tightened the peg under normal conditions.

DAI is unique in the dataset in that it depegs in both directions. During risk-off environments it often trades above the dollar as demand for decentralized stablecoins rises faster than new supply can be minted. During liquidation cascades it falls below the dollar as collateral values drop faster than the protocol can auction positions. The March 2020 Black Thursday episode was DAI's worst, with the price reaching approximately $0.891 and a maximum deviation of around 11% sustained for approximately 96 hours. The mechanism was a cascade: ETH fell roughly 50% in under 24 hours, triggering mass liquidations of MakerDAO vaults. Network congestion on Ethereum caused liquidation auctions to clear at effectively $0 per collateral unit, leaving the system under-collateralized until an emergency MKR token dilution stabilized it. The March 2023 SVB contagion produced a comparable 10.5% maximum deviation over 72 hours, driven through a different channel: DAI holds substantial USDC as collateral through the PSM, so when USDC depegged, DAI's effective backing value declined simultaneously.

The predictive signal structure for DAI differs from the fiat-backed coins in one critical respect: Curve pool features rank significantly higher. The Curve 3pool imbalance is DAI's primary arbitrage mechanism — when the pool tilts heavily toward DAI, it signals that sellers are exiting into USDC and USDT faster than arbitrageurs can rebalance. This feature is largely irrelevant for USDT because USDT does not rely on Curve as its dominant exit route. ETH return is also a strong signal for DAI, acting as the principal collateral stress driver unique to the crypto-backed architecture.

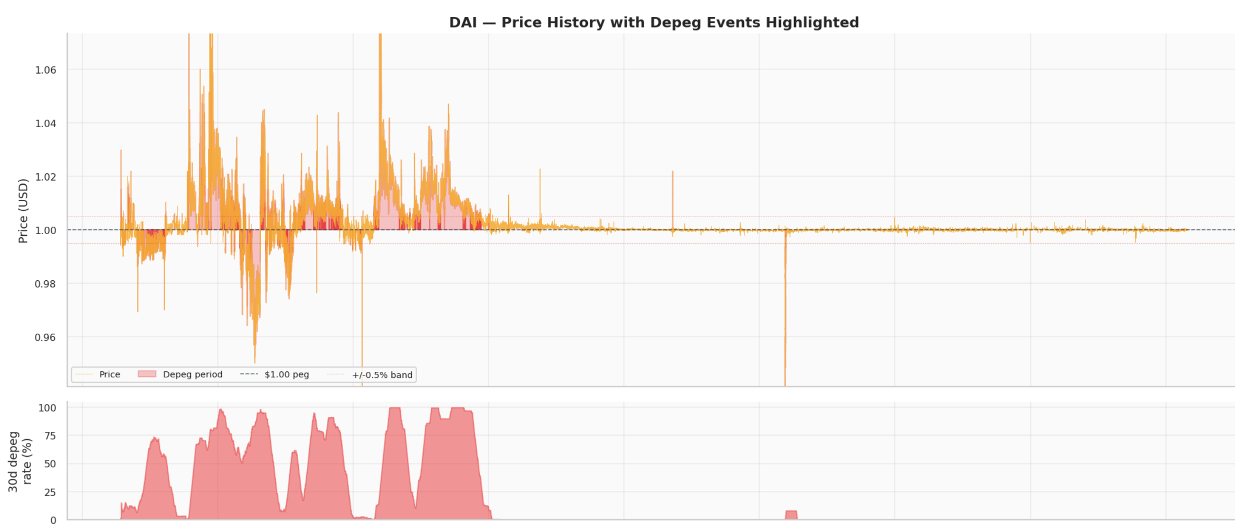

Figure 3. DAI Pricing History

BUSD — Binance USD (Fiat-backed, 2019–2023)

BUSD has 370,848 rows covering September 2019 to March 2023, with the dataset ending after the Paxos enforcement action. Its pre-FTX history is among the cleanest in the dataset, with a normal operating range of plus or minus 0.05% and a depeg rate of only 0.3 to 0.8% across its active life. The coin's risk profile is therefore essentially binary: essentially no stress for three years, then a single concentrated event. The November 2022 FTX collapse produced BUSD's sole Severe episode, with a minimum price of approximately $0.963 and a maximum deviation of 3.7% lasting around 48 hours. The mechanism was a compounding of three simultaneous shocks: exchange counterparty risk as FTX halted withdrawals, Binance's public announcement that it was exiting its BUSD position, and the beginning of NYDFS regulatory scrutiny.

The consequence of this single-event history for the predictive analysis is that macro and sentiment features dominate over on-chain flows, unlike the pattern seen in USDT and USDC. This reflects the news-shock mechanism: BUSD's depeg was triggered by information rather than by gradually building redemption pressure, and sentiment indicators were the fastest channels through which that information was priced. The relative importance of macro signals — Fear and Greed and BTC return — is notably higher for BUSD than for its fiat-backed peers.

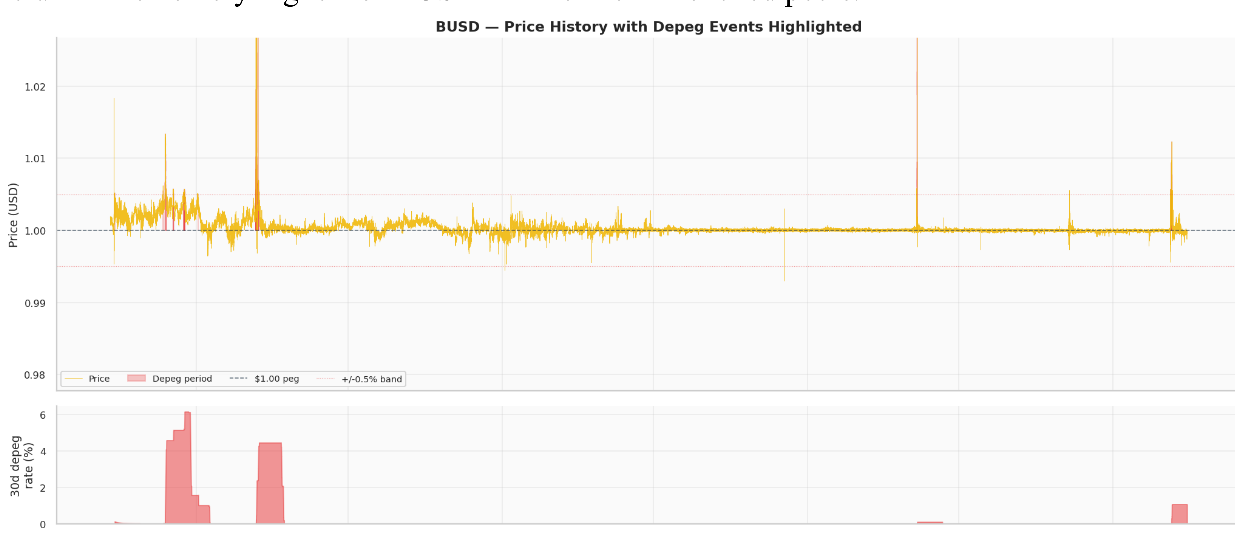

Figure 4. BUSD Pricing History

UST — TerraUSD (Algorithmic, 2020–2022)

UST is the most analytically unusual entry in the dataset. Its 153,961 rows cover November 2020 to May 2022, and the coin effectively ceased to exist as a functioning stablecoin on May 9, 2022. The depeg rate is approximately 100% for all rows after that date, as the price fell from its peg to essentially zero and never recovered. This means UST must be treated as two fundamentally separate regimes: a stable pre-collapse window and a post-collapse window that represents a permanent, total failure rather than a recoverable depeg event.

The algorithmic mechanism made UST uniquely fragile. The peg was maintained entirely by an arbitrage relationship with LUNA — users could burn $1 worth of LUNA to mint 1 UST, or burn 1 UST to receive $1 worth of LUNA. This mechanism is self-stabilizing in normal conditions but catastrophically unstable under large sustained selling pressure: selling UST to mint LUNA increases LUNA supply, which depresses LUNA price, which means each subsequent redemption mints more LUNA than the last, accelerating the decline of both assets simultaneously. The May 2022 collapse began with a coordinated Anchor protocol withdrawal of approximately $14 billion in TVL, depleted the Luna Foundation Guard reserve that was the only external buffer, broke the mint/burn arbitrage mechanism, and triggered LUNA hyperinflation within approximately 72 hours.

The most practically important finding from the UST predictive analysis is the inversion of the mint signal. For USDT and USDC, high mint volumes indicate demand and confidence, while high burn volumes signal stress. For UST, the opposite is true: elevated minting in the pre-collapse window reflected protocol defense attempts, making it a stress indicator. Any model pooling UST with fiat-backed coins without a coin-type flag will be misled by this signal. The severity transition matrix for UST once Severe status is reached shows a self-transition probability approaching 100%, the mathematical signature of an irreversible death spiral that is unlike any other coin in the dataset.

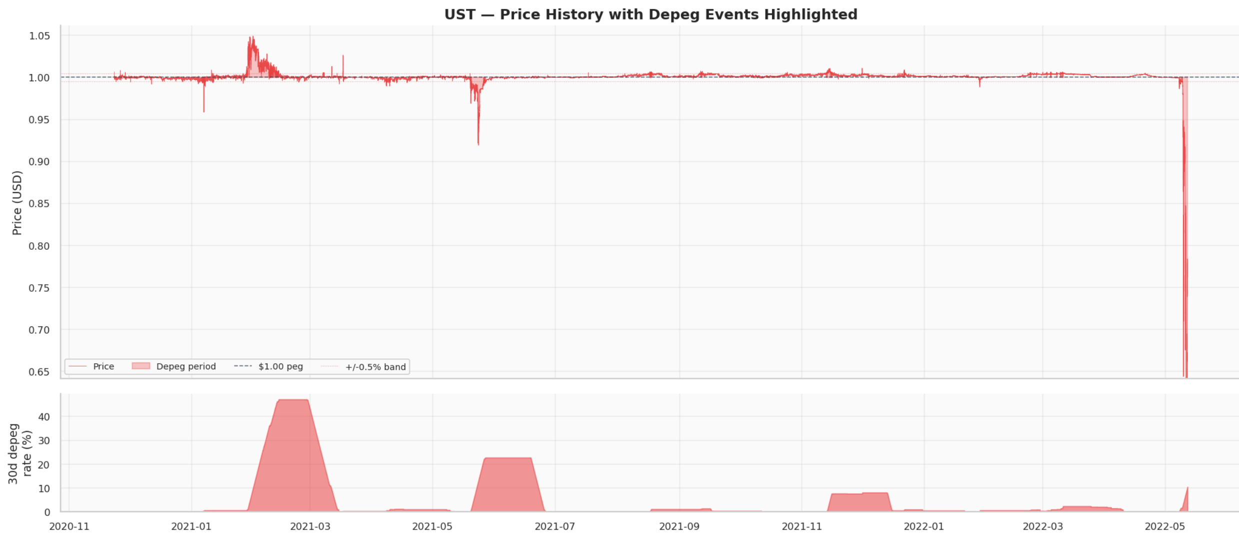

Figure 5. UST Pricing History

USDe — Ethena (Yield-bearing, 2024–2026)

USDe is the newest coin type in the dataset, with 200,928 rows covering April 2024 to February 2026. The peg is maintained through delta-neutral derivatives positions on centralized exchanges, with yield from those positions distributed to holders. This means USDe's risk profile is tied to derivatives market conditions rather than to reserve confidence, collateral values, or algorithmic relationships. Under normal conditions it is among the most tightly pegged coins in the dataset with a range of plus or minus 0.05 to 0.10%. Its single observed stress event occurred in August 2024 when BTC fell approximately 20% and ETH approximately 25%, driving funding rates deeply negative. The Ethena insurance fund absorbed the shortfall and the coin recovered, with a maximum deviation of approximately 2.1% classified as Moderate. BTC and ETH returns dominate the predictive signal rankings, with on-chain flow features significantly less informative than for the reserve-based coins.

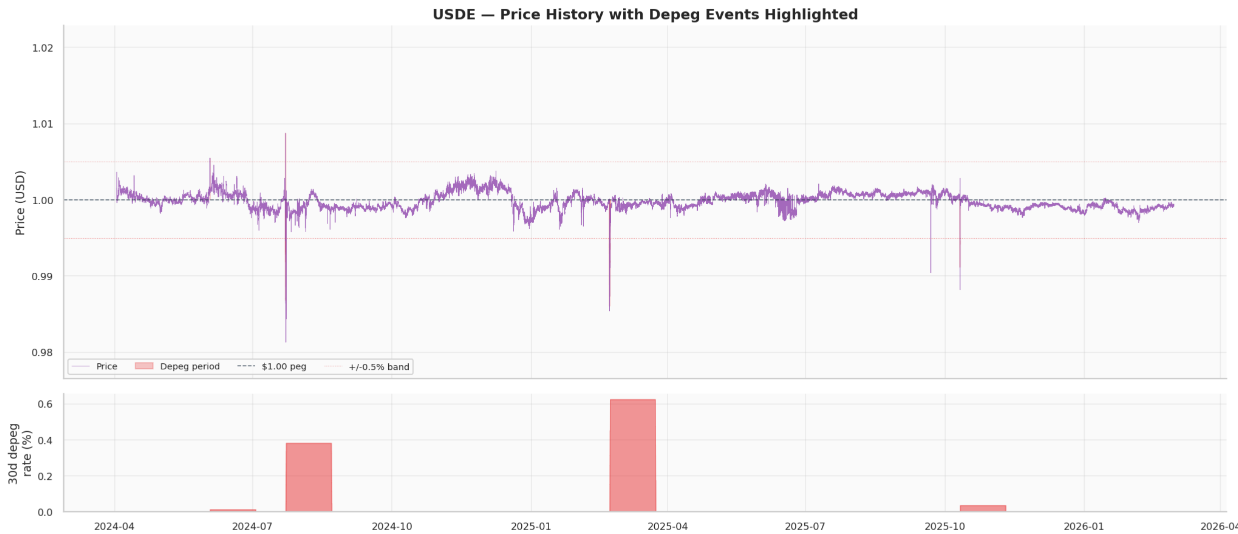

Figure 6. USDe Pricing History

RLUSD — Ripple USD (Fiat-backed, 2025–2026)

RLUSD has 96,192 rows covering April 2025 to February 2026. It is the cleanest coin in the dataset with a normal operating range of plus or minus 0.02 to 0.05%, a maximum observed deviation below 0.2%, and a depeg rate of approximately 0%. No depeg episodes have been observed, meaning no severity classification is possible and the AUC analysis cannot be run. RLUSD serves two purposes: as a clean-peg baseline reference, and as a contributor to the cross-coin peer deviation features, where a reading near zero confirms that stress being detected in another coin is idiosyncratic rather than systemic. Feature selection should be revisited once at least one crisis event is observed.

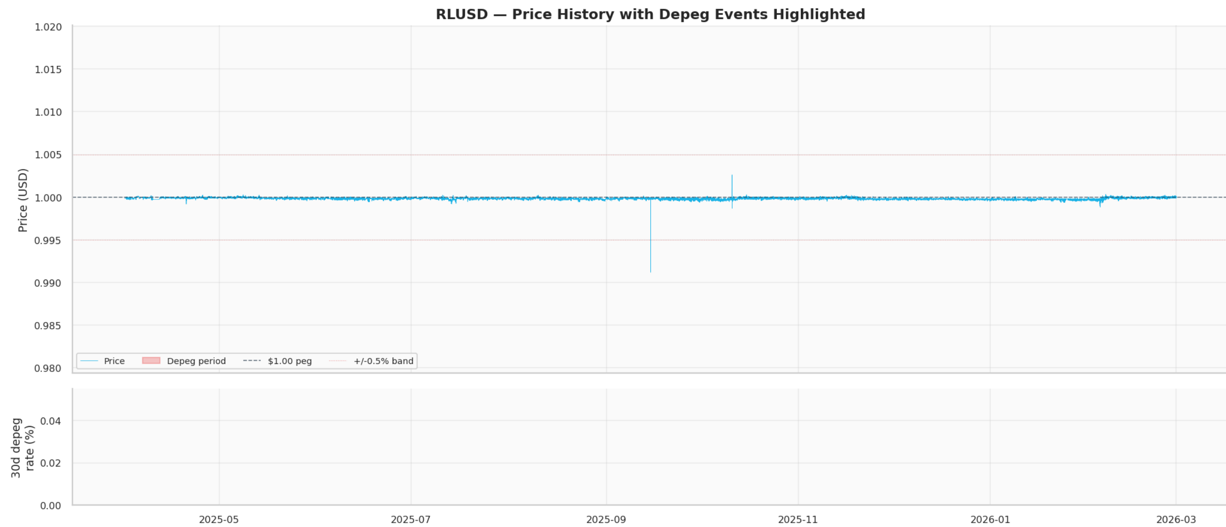

Figure 7. RLUSD Pricing History

Cross-Coin Analysis

When the seven coins are examined together, two structural patterns emerge that are not visible within any single coin's data: a clear hierarchy of depeg risk by coin type, and a pattern of contagion co-occurrence that reveals the channels through which stress propagates across the stablecoin ecosystem.

The most fundamental finding from the cross-coin comparison is that depeg rate is largely determined by mechanism type. Algorithmic coins carry structurally unlimited downside — UST reached effectively 100% depeg with no recovery. Crypto-backed coins face ongoing collateral volatility risk, giving DAI the highest sustained depeg rate among surviving coins. Fiat-backed coins have the lowest typical depeg rates, with depegs occurring only during discrete external shocks. Yield-bearing coins sit in a similar range to fiat-backed coins under current conditions but face a different stress channel through derivatives market dislocations. This hierarchy has direct implications for model design: a single pooled model treating all coins identically will be systematically mis calibrated unless coin mechanism type is included as a feature, which the final 20-feature set addresses directly.

The Depth by Duration severity matrix assigns one of four tiers — Normal, Mild, Moderate, or Severe — to each depeg episode. Episodes with a maximum deviation below 0.5% are always Mild. Episodes between 0.5% and 2% are Mild under six hours and Moderate for longer durations. Episodes between 2% and 10% are Moderate under six hours and Severe for longer. Any episode with a maximum deviation above 10% is Severe regardless of duration — the hard override that captures UST's collapse and USDC's SVB episode. Across all coins combined, most episodes are Mild, with Moderate and Severe episodes concentrated in the six crisis windows. The severity transition matrices confirm that for all coins except UST, depegs are recoverable — the probability of remaining Severe at the next five-minute bar is substantially below 100%. UST's matrix is the exception, with near-certain self-transition once Severe is reached.

The depeg co-occurrence analysis measures the percentage of overlapping time periods where both members of a coin pair were simultaneously depegged. USDT and USDC have the highest co-occurrence, reflecting their shared exposure to the 2022 crisis cluster. DAI and USDC show elevated co-occurrence due to the direct collateral channel: DAI holds USDC through the PSM, so a USDC depeg directly reduces DAI's effective backing, as demonstrated in March 2023. UST shows co-occurrence with USDT and USDC only during the May 2022 window, consistent with the systemic shock that briefly destabilized the entire ecosystem. BUSD and RLUSD show near-zero co-occurrence with all others, reflecting their single-event and zero-event risk profiles respectively. Two systemic spike windows are confirmed in the cross-coin rolling timeline: May 2022, when UST, USDT, USDC, and DAI all elevated simultaneously, and November 2022, when BUSD spiked in isolation.

Aggregating univariate AUC results across all coins reveals a consistent signal hierarchy despite the mechanistic differences between coin types. Price deviation rolling statistics are the strongest signals across every coin for which sufficient depeg history exists, with one-hour rolling standard deviation achieving adjusted AUC values between 0.75 and 0.95. The consistent leadership of this group reflects a common dynamic: volatility rises before the formal depeg threshold is crossed, providing a brief window of advance warning even in news-driven events.

CEX microstructure signals rank second but are available only for USDT. On-chain flow signals from mint and burn volumes rank third overall, with the important caveat that the signal direction is inverted for UST. Curve DEX pool features are more informative for DAI and USDC because Curve is the primary arbitrage venue for those coins. Cross-market signals from BTC and ETH returns provide regime context most relevant for UST collapse prediction and USDe stress, where the mechanism runs directly through crypto price dynamics. Macro and sentiment indicators from VIX and Fear and Greed add lower-frequency regime context that is most useful in combination with price and flow signals rather than in isolation.

Feature Engineering

The feature engineering pipeline processes seven cleansed per-coin parquets and produces a single pooled file containing only onset rows — bars where the coin is not currently depegged — to focus the model on predicting depeg initiation rather than continuation. Engineering is performed per coin before pooling to prevent z-score contamination across coins with different date ranges and volatility profiles. The chronological split places 70% of data in training through March 2024, with 15% validation and 15% held back as a test set for Notebook 3, yielding 3,029,653 pooled rows with a depeg_next_1h positive rate of 0.94%.

The 46 engineered features divide into three categories. Universal features, available for all coins, number 23 and include signed and absolute price deviation in basis points, one-hour and six-hour rolling volatility, six-hour momentum, a seven-day z-score, intrabar high-low range, trading volume surge and z-score measures, BTC and ETH 24-hour returns and volatility, VIX and DXY macro indicators, the Fear and Greed index, cross-coin price deviations for USDT, USDC, and DAI peers, maximum peer deviation, and the hour and day-of-week temporal features. Supplementary features, available only for specific coin subsets, number 19 and cover Binance

CEX microstructure for USDT, Curve pool net-sell pressure and lags for coins with 3pool access, and on-chain net flow z-scores and lags for coins with mint/burn data. Structural absence flags, always binary and never NaN, number four and make the absence of CEX data, Curve pool access, a coin's own DEX, and on-chain flow data explicit rather than encoding it as zero, which would mislead the model.

Because USDT and DAI together dominate the row count, episode-proportional sample weights are computed and passed to the model. Each coin's weight equals its share of total depeg episodes divided by its share of total rows, correcting for USDT and DAI row dominance and substantially upweighting DAI. RLUSD receives a weight of zero because it has no positive-class observations. This weighting has a measurable positive effect on validation AUC for minority coins.

Feature selection uses CatBoost's recursive elimination by loss function change on a 200,000-row subsample. Three thresholds are tested: 15 features producing a validation AUC of 0.9730, 20 features producing 0.9764, and 25 features producing 0.9595. The 20-feature threshold is selected as the optimum, representing the point where each additional feature adds meaningful signal before diminishing returns and noise begin to dominate. The 20 selected features span all major signal groups. From price deviation they include signed deviation, absolute deviation, one-hour volatility, six-hour volatility, and intrabar high-low range. From volume they include the six-hour trading volume surge and the seven-day volume anomaly z-score. From the cross-market group they include BTC 24-hour return, ETH 24-hour return, and BTC 24-hour volatility. From macro they include VIX and the Fear and Greed index. From cross-coin they include USDC peer price deviation. From temporal features the day of week is included. The coin mechanism type categorical variable and the on-chain flow structural flag serve as architectural inputs. From supplementary on-chain features the net flow z-score versus seven days, the 24-hour net flow total, net flow lags at 15 minutes and one hour, and the Curve pool sell pressure z-score over 30 days are included.

Several features that might have been expected to make the final set were excluded on the evidence. DXY and Federal Funds rate are redundant once BTC return and VIX are included, because the crypto-native signals already capture the macro regime information those indicators provide. Curve DEX lag features are subsumed by the sell pressure velocity feature. Net flow lags beyond the 15-minute and one-hour horizon add little once the z-score normalization accounts for recent baselines. Hour of day has a low standalone AUC and lost competition to on-chain and market features for the available slots. The final model's validation AUC of 0.9764 confirms that on-chain flow features and price volatility statistics are the backbone of the predictive signal, with macro and cross-coin features providing meaningful but secondary contributions. The LOEO holdout evaluation in Notebook 3 will determine whether this generalization holds across the six structurally distinct crisis event types not seen during training.

Modelling

The modelling stage operationalizes the features engineered in the previous section into a deployable early warning system. Our design objective is twofold:

1. Detect stablecoin depeg events with sufficient lead time to inform defensive action.
2. Remain explainable and auditable so risk analysts and regulators can trust the output.

Stablecoin depegs are rare events that unfold on compressed timelines, which places unusual demands on the modelling framework. The choice of algorithm, validation scheme, and threshold policy reflects these constraints.

Architecture Overview

The final system comprises two models working in concert. A supervised CatBoost classifier addresses the primary warning horizon of one hour, and an Isolation Forest serves as an unsupervised novelty detector for regimes that fall outside the labelled depeg patterns seen during training.

- **CatBoost (depeg_next_1h).** A gradient-boosted classifier that predicts whether a depeg event will occur within the next 12 five-minute bars. This is the system's primary horizon and drives the high-risk tier.
- **Isolation Forest.** An unsupervised detector trained exclusively on rows that were stable both in the current bar and the subsequent 30 minutes. Flags observations that are statistically abnormal relative to learned normal behavior, regardless of whether the shape of that abnormality matches any known historical crisis.

This two-layer design separates concerns. The supervised model exploits the full labelled history of depegs to answer the question "does this bar look like the precursors of past depegs?". The Isolation Forest answers a different question: "does this bar look normal at all?". A crisis that rhymes with past events is caught by CatBoost. A crisis that is structurally novel is caught by the Isolation Forest. Together they cover both known and unknown unknowns.

Supervised model Algorithm - CatBoost

CatBoost was chosen as the supervised algorithm after evaluating nine candidate architectures, including XGBoost, LightGBM, Random Forest, Logistic Regression, LSTM, LSTM Autoencoder, Gaussian Mixture Model, and One-Class SVM. CatBoost's selection was driven by three properties well-suited to the problem. First, its ordered boosting algorithm mitigates target leakage when training on time-series features by computing target statistics on permutation-ordered subsets rather than the full training set, ensuring predictions use only information that would have been available at or before that point in time. Second, CatBoost handles missing values natively without imputation, which is essential for our feature set because supplementary on-chain and DEX pool features are only available for a subset of coins and periods. Third, CatBoost provides built-in class imbalance handling via the auto_class_weights='Balanced' parameter, supplemented with row-level sample weighting that allows our episode-proportional coin weights to apply directly to the loss function.

XGBoost and LightGBM were retained as documented alternatives. Both matched CatBoost on overall AUC-PR, but neither offers native ordered boosting and both would require additional tuning to prevent temporal leakage. The remaining six candidates were rejected for the following reasons. Random Forest and Logistic Regression do not handle NaN natively and lack the expressiveness for highly non-linear decision boundaries, the deep learning architectures (LSTM, LSTM Autoencoder) were unstable on our small positive class count and did not converge reliably in cross-validation, Gaussian and One-Class SVM are unsupervised and could not exploit the labelled depeg history at all.

Feature Set

Twenty model inputs are passed to CatBoost: 19 numeric features and one categorical feature (u_coin_type). Features are named with prefixes that encode their provenance and interpretation. Universal features (prefix u_) are derived from price or return data available for all seven coins across the full study period, and include u_volatility_6h, u_abs_dev_bps, u_volatility_1h, u_eth_return_24h, u_btc_return_24h, u_fear_greed, and u_day_of_week. Supplementary features (prefix s_) are derived from on-chain or DEX pool data only available for coins and periods where such data was collectible, including s_pool_net_sell_z_score_30d, s_net_flow_z_score_7d, s_net_flow_24h, s_net_flow_lag3, and s_net_flow_lag12. Flag features (prefix f_) are binary indicators marking the presence or absence of supplementary data, such as f_has_onchain, and allow the model to distinguish between "supplementary feature is zero" and "supplementary feature is missing", which is semantically different and materially affects the prediction. The categorical u_coin_type allows the model to learn coin-specific patterns directly. For instance, BUSD exhibited different pre-depeg dynamics than DAI, and including coin identity as an input lets the model account for these differences within a single unified model rather than requiring a separate model per coin.

Data Split Design

We use a coin-stratified chronological 70/15/15 split into training, validation, and test sets. Each coin's own history is split chronologically within itself, and the per-coin slices are pooled into the three sets. This ensures every coin contributes rows to all three splits while preserving temporal ordering within each coin. Ordering is critical. A random split would leak future information into training. A global chronological split without stratification would silently exclude short-lived coins such as UST, whose history runs only from November 2020 to May 2022 and falls entirely within the training window of a global split.

After splitting, the training set contains 2,120,754 rows, validation contains 454,449 rows, and test contains 454,450 rows. The test set is held out and touched exactly once at the end of the modeling process. All hyperparameters, thresholds, and calibration fits are determined using only training and validation data.

Class Imbalance Handling

The depeg class is severely imbalanced — depeg_next_1h has a 0.944% positive rate across the training set. Three complementary mechanisms address this imbalance. First, CatBoost's auto_class_weights='Balanced' setting weights the loss contribution of each class inversely to its training frequency, equivalent to up sampling the minority class without the memory cost of doing so explicitly. Second, we apply episode-proportional sample weights at the row level. Each row carries a coin-specific scalar w_coin equal to the ratio of that coin's share of crisis episodes to its share of total rows. DAI receives the highest weight (2.2798) because its Black Thursday episode is dense with labelled positives; RLUSD receives a weight of zero because it has no confirmed depeg events in the study period. The full weight schedule is USDT 0.9712, USDC 0.4163, DAI 2.2798, BUSD 0.1277, UST 1.1562, USDE 0.0707, RLUSD 0.0000. Third, we report AUC-PR (area under the precision-recall curve) as the primary evaluation metric rather than AUC-ROC, because at such low positive class rates ROC curves are misleading — a model that achieves AUC-ROC of 0.99 can still have precision below 10% at any operating threshold. The random-baseline AUC-PR on the validation set is 0.0002, so even an AUC-PR of 0.10 would be a meaningful improvement.

Hyperparameters

CatBoost is configured with a base hyperparameter set that was fixed early and validated to perform well across multiple trial runs. We deliberately avoid extensive hyperparameter tuning because the validation set contains only a few hundred positive samples. Aggressive tuning would likely overfit to that specific validation window. The configuration is shown below

Hyperparameter	Base Training	LOEO Retraining	Rationale
iterations	500	200	Early stopping typically halts at ~20–200 rounds
depth	6	5	Moderate depth balances expressiveness against overfitting on rare events
learning_rate	0.05	0.05	Conservative, paired with early stopping
loss_function	Logloss	Logloss	Standard binary classification loss
eval_metric	AUC	AUC	AUC used for early stopping; AUC-PR used for final evaluation
auto_class_weights	Balanced	Balanced	Inverse-frequency class weighting

early_stopping_rounds	50	50	Stops if validation AUC does not improve for 50 rounds
random_seed	42	42	Reproducibility
max_ctr_complexity	default	1	Reduced for LOEO to speed up retraining on each event

Table 1. *CatBoost hyperparameter configuration for base training and LOEO cross-validation.*

The model converged at iteration 188 with validation AUC-ROC of 0.9996 and validation AUC-PR of 0.4965.

Probability Calibration

CatBoost outputs raw probabilities that are not well calibrated when class weights are used in training, because the model effectively minimizes a weighted loss on an up sampled representation of the positive class. Raw probabilities overestimate the positive class rate at every threshold. We apply Platt scaling to correct this, where a logistic regression is fit on the validation set using the model's raw probabilities as the single input feature and the true label as the target, and the resulting sigmoid is applied to future predictions. Calibration matters for threshold selection — using raw probabilities would require re-tuning the threshold per model run, whereas calibrated probabilities allow thresholds to be interpreted as "probability of depeg" and remain stable across retrains.

Threshold Optimisation

After calibration, we select the operating threshold by maximising the F2 score on the validation set. F2 is a generalisation of the F1 score that places twice as much weight on recall as on precision. The choice of F2 over F1 encodes the cost asymmetry of this problem. Missing a real depeg can cost billions of dollars at the protocol level and tens of thousands at the individual trader level, whereas a false alarm costs at most one unnecessary defensive action, whether that is premature exit, a re-hedge, or an analyst hour spent confirming the signal is spurious. Sweeping threshold over a grid of 100 values between 0.05 and 0.50, we find that the F2 maximum of 0.6338 occurs at a threshold of 0.1273, shown in Figure 1.

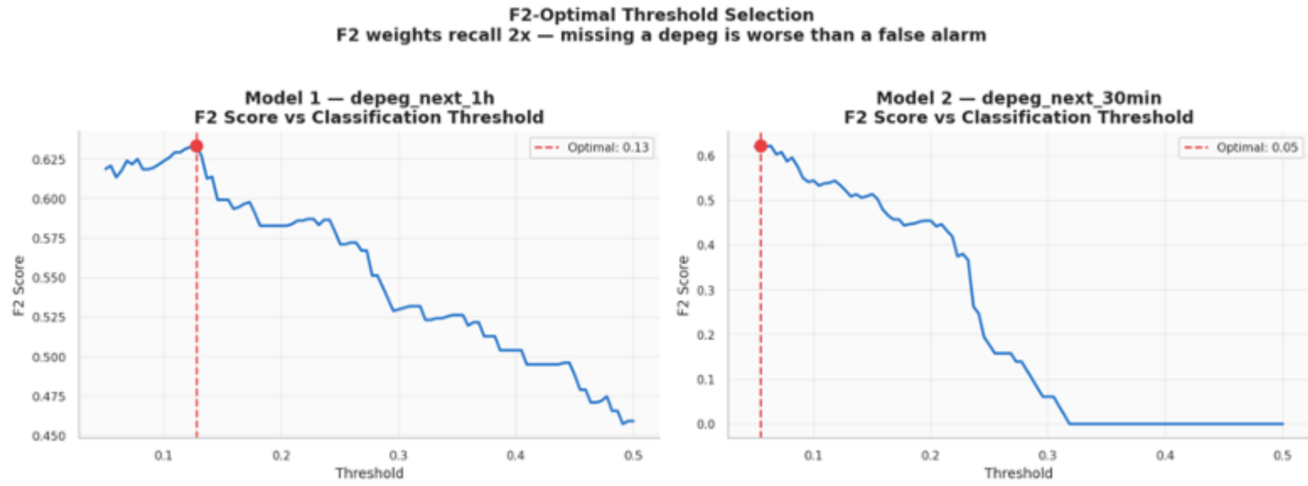

Figure 8. *F2 score as a function of classification threshold. The dashed red line marks the F2-optimal threshold of 0.13. F2 degrades sharply above the optimum because precision gains cannot compensate for the recall penalty.*

The precision-recall curve on the validation set, shown in Figure 9, illustrates the full set of trade-offs available to the practitioner. The red dot marks the operating point selected by F2 optimization.

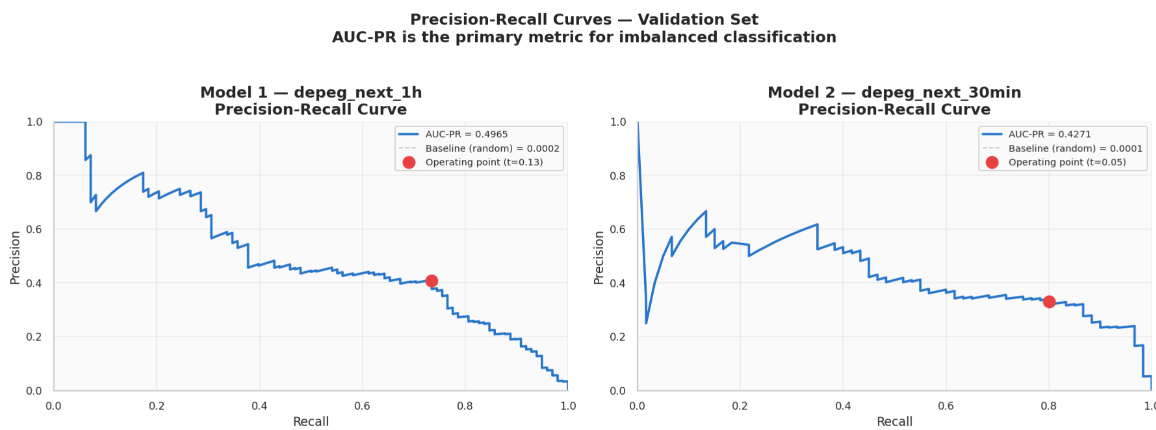

Figure 9. *Precision-recall curve on the validation set. The grey dashed line at the bottom is the random baseline (equal to the positive class rate of 0.02%). The red marker is the F2-optimal operating point. AUC-PR of 0.4965 is approximately 2,480× better than random.*

Leave-One-Event-Out Cross-Validation

Held-out performance on a chronological test set is necessary but not sufficient for an early warning system. Test-set metrics tell us how the model performs on a mix of normal periods and crisis episodes drawn from the most recent 15% of each coin's history, but they can be dominated by performance on the normal periods. The question that really matters is whether the model would have fired on a crisis it had never seen before. This is the question that Leave-One-Event-Out (LOEO) cross-validation answers.

For each of five confirmed historical depeg events, we define an event window and a surrounding 48-hour buffer. The buffer prevents leakage through rolling features. A feature computed over the 6-hour window ending at a point inside the buffer would overlap the event itself, so including any buffer row in training would give the model partial information about the event it is supposed to predict out-of-sample. We then construct a training set that excludes every row falling inside the event-plus-buffer, retrain a CatBoost model from scratch using the LOEO_PARAMS configuration, and evaluate on the event window itself.

The five events are: the Oct 2018 USDT Reserve Scare, the Mar 2020 DAI Black Thursday depeg, the May 2022 UST algorithmic collapse, the May 2022 USDC contagion depeg caused by UST spillover, and the Mar 2023 USDC SVB Bank Run depeg. Two other candidate events (BUSD/FTX and a USDE selloff) were excluded because their labelled positive counts were too low (under 35 bars) to yield stable evaluation metrics.

We compute two metrics per event. The first is AUC-PR on the event window, which measures how well the model ranks event rows ahead of non-event rows. The second is lead time, the number of hours before the first labelled depeg bar at which the model first fires at or above its F2-optimal threshold. The results are shown in Table 2 and Figure 3.

Event	Depeg Bars	AUC-PR	Lead Time (hrs)
Oct 2018 USDT Reserve Scare	15	1.0000	0.0
Mar 2020 Black Thursday (DAI)	67	0.6631	23.2
May 2022 UST Collapse	71	0.5577	0.0
May 2022 UST Contagion (USDC)	12	0.0664	15.9
Mar 2023 SVB Bank Run (USDC)	37	0.7201	0.0

Table 2. *LOEO cross-validation results for the five confirmed historical depeg events. Lead time is the number of hours before the first labelled depeg bar at which the model first fired at or above its F2-optimal threshold. A lead time of 0.0 indicates the model fired on the first depeg bar (a coincident detection rather than an early one).*

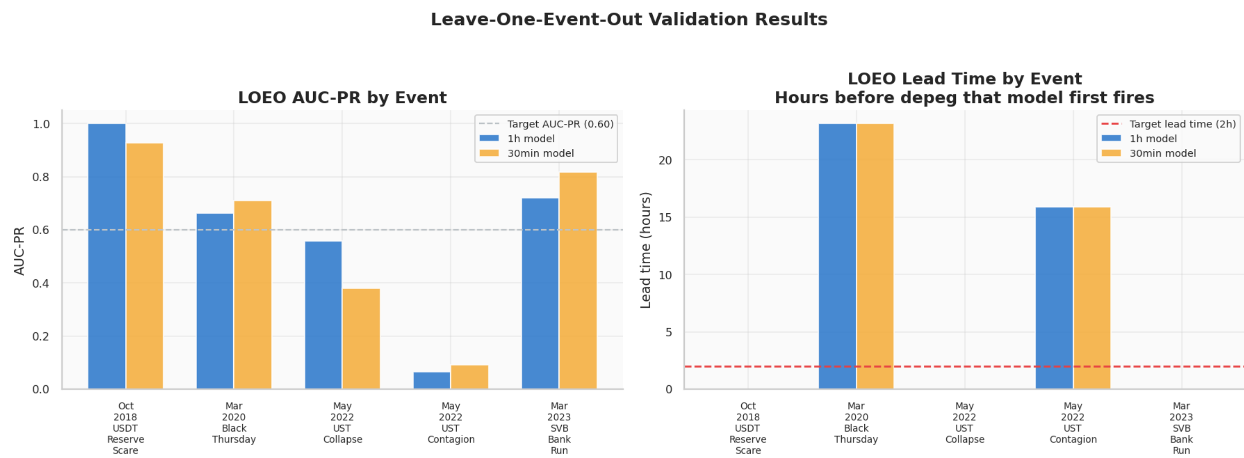

Figure 10. *LOEO validation results by event. Left panel: AUC-PR by event, with a target line at 0.60. Right panel: lead time in hours, with a target line at 2 hours. Black Thursday (DAI, Mar 2020) and the UST Contagion event (USDC, May 2022) achieved the target lead time. The USDT 2018 event, UST collapse, and SVB bank run were coincident detections with zero lead time.*

The results exhibit a clear dichotomy. Black Thursday and the UST Contagion event were detected well in advance (23.2 and 15.9 hours respectively). The USDT 2018 Reserve Scare, the UST collapse itself, and the SVB bank run were coincident detections — the model fired on the first depeg bar rather than before it. The lead time outcome is largely a function of the mechanics of each event. Depegs driven by on-chain pressure that builds over hours leave pre-event fingerprints in the 6-hour volatility, 30-day pool sell pressure, and 7-day net flow features that the model learns to recognize. Depegs driven by sudden exogenous shocks arrive without precursors the features can see — the price break is itself the first signal. The model can still detect these coincidentally, which gives it value as a confirmation mechanism even when it cannot provide forward warning.

Unsupervised model algorithm - Isolation Forest

The Isolation Forest is trained on a restricted subset of the data. It learns from rows where the coin is stable at the current bar and will remain stable for at least the next 30 minutes. This "confirmed normal" definition is stricter than simply using non-depeg rows, because it excludes rows that are themselves in the run-up to a depeg but have not yet crossed the depeg threshold. Training on confirmed normal rows ensures the Isolation Forest's baseline is genuinely representative of calm market behaviour. A total of 3,001,053 rows satisfy this definition, representing 99.1% of the full dataset. Input features are restricted to the 11 universal numeric features, because Isolation Forest treats NaN as a regular feature value rather than routing it through a learned branch, and our supplementary features have systematic missingness that would dominate the anomaly score.

The model outputs an anomaly score ranging from 0 (maximally normal) to 100 (maximally anomalous). This score is derived from the Isolation Forest's raw score via a percentile-rank transformation computed on the training distribution, which makes it interpretable as how rare an observation is relative to normal market history. An alert threshold of 80 was selected based on the score distribution. Scores above 80 correspond to the top 0.28% of observations, which produces a manageable alert volume for a monitoring desk. Figure 4 shows both the distribution and the time series of the rolling 1-hour average anomaly score.

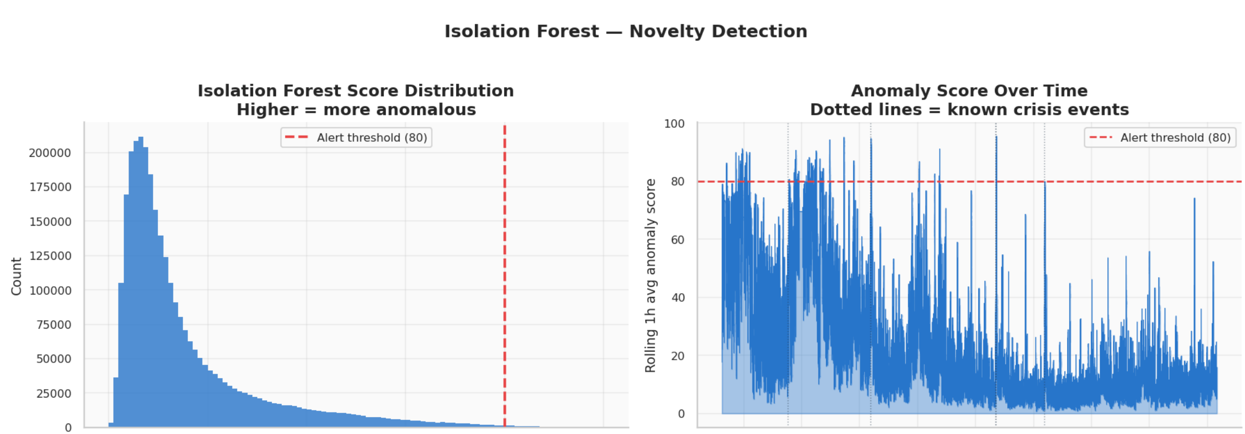

Figure 11. *Isolation Forest diagnostics. Left: distribution of anomaly scores across the full 3.0M-row dataset. The alert threshold is set at 80, above which 0.28% of observations fall. Right: rolling 1-hour average anomaly score over the full study period, with vertical dotted lines marking the five confirmed crisis events.*

The time series panel reveals two important properties. First, known crisis events produce clear spikes in the rolling anomaly score, validating that the Isolation Forest recognizes them as statistically abnormal. Second, there are other anomaly spikes in the time series that do not coincide with labelled events. These represent either mislabeled near-misses, transient liquidity events that did not develop into full depegs, or genuine novel stress regimes that deserve investigation. The Isolation Forest's value as a novelty layer is precisely that it surfaces these periods; when both CatBoost and the Isolation Forest fire simultaneously, the alert carries maximum confidence.

Final Test Set Performance

The test set is evaluated exactly once after all modelling choices are locked. On the 454,450-row test set, CatBoost achieves AUC-PR of 0.4494, AUC-ROC of 0.9985, F2 of 0.5798, precision of 0.349, and recall of 0.695. Precision of 0.349 means that out of every 100 alerts fired, roughly 35 correspond to actual future depegs. This represents a lift of approximately 37× over the 0.944% base rate.

Limitations

Class Imbalance

Depeg events are rare by design — the pooled dataset is ~9% positive across all coins, and true severe depeg episodes are far rarer still. Despite sample weighting and CatBoost's auto_class_weights, the model is biased toward predicting stability. Precision at operationally useful thresholds remains low, meaning alert systems built on these predictions would generate a significant number of false positives.

Missing Order Book Data

Order book snapshots were not available during this project. Order book depth and bid-ask spread are among the strongest real-time signals of institutional stress — a thinning order book on the sell side often precedes price dislocations. Without this data, the model relies on lagged price and volume signals rather than the forward-looking liquidity information that order book snapshots would provide. This is the most significant gap in the feature set.

Survivorship and Recency Bias

Only 7 coins are included, and major depeg events are concentrated in 2022–2023 (UST collapse, SVB/USDC). The model has limited exposure to diverse depeg mechanisms and may not generalize to novel stablecoin designs or stress regimes not seen in the training window.

Findings

The modelling and validation exercise yields findings at two levels. First, at the feature level, we identify which market, on-chain, and macro signals contribute the most predictive power and in which direction. Second, at the system level, we characterize when the model is most and least effective and what this implies about the nature of stablecoin depeg risk. Both sets of findings are grounded in the LOEO results and the SHAP decomposition of the final model.

Key Indicators of Stablecoin Stress

SHAP (Shapley additive explanations) values provide a principled decomposition of each individual prediction into contributions from each feature. Aggregated across the 5,000-row test-set sample on which SHAP was computed, the mean absolute SHAP value per feature ranks the features by their total influence on model output. Figure 12 shows this ranking for the full 19-feature set.

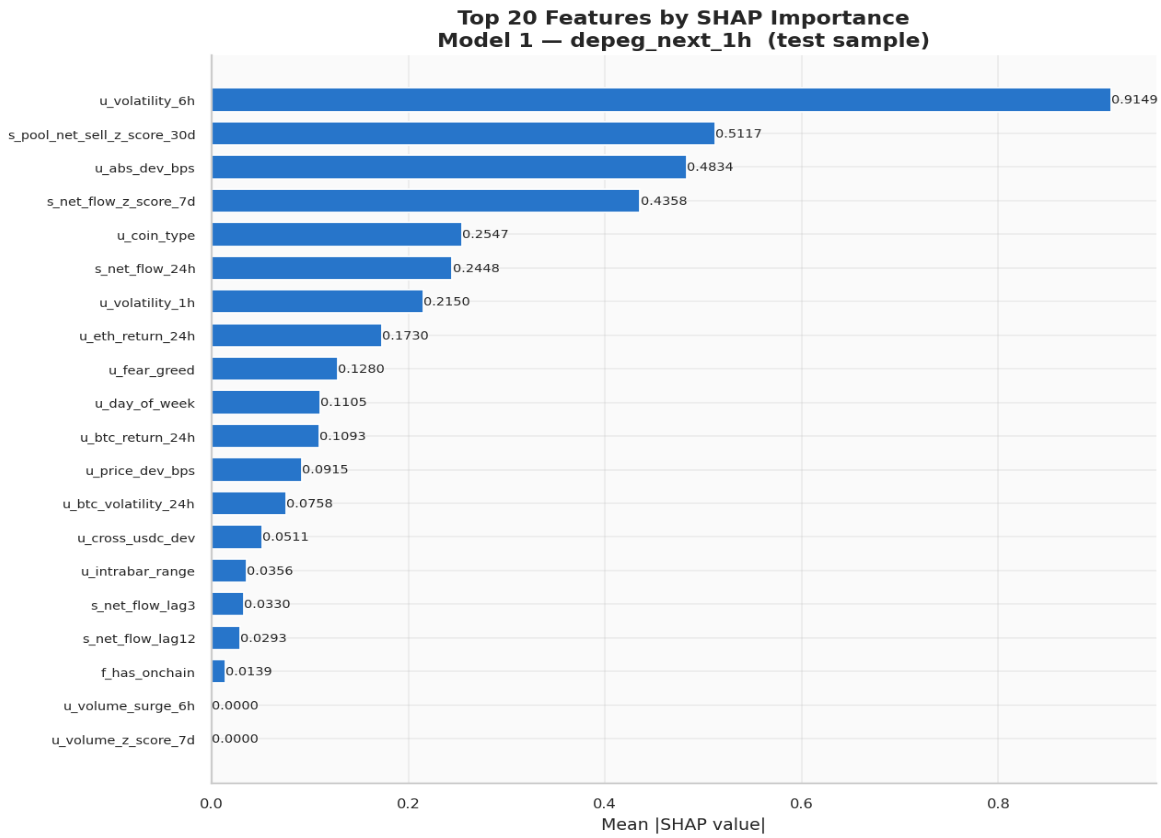

Figure 12. *Mean absolute SHAP value per feature for the CatBoost model, computed on a 5,000-row test-set sample. Six-hour volatility dominates the ranking, nearly double the second-ranked feature.*

Five features carry the majority of the predictive weight. Their combined mean absolute SHAP accounts for approximately 80% of the total feature importance mass. In decreasing order:

- **u_volatility_6h (mean |SHAP| = 0.9149).** Six-hour price return volatility. This is the single most important feature by a substantial margin, nearly double the second-ranked feature. High six-hour volatility pushes the prediction strongly toward depeg. Volatility is a near-universal precursor signal: before a coin breaks peg, it typically oscillates within a narrowing band relative to its normal jitter, and this feature captures that elevated dispersion directly.

- **s_pool_net_sell_z_score_30d (mean |SHAP| = 0.5117).** The 30-day z-score of net sell pressure in the coin's primary DEX liquidity pool. High sustained sell pressure over a month-long window indicates accumulating redemption demand that has not yet been met by offsetting arbitrage or new minting. This feature is available only for DAI and USDC (which have large Curve pools); for other coins it is NaN, and CatBoost routes NaN to a default branch.
- **u_abs_dev_bps (mean |SHAP| = 0.4834).** The absolute deviation of the current price from the $1.00 peg, in basis points. This feature is almost tautologically predictive: a coin that is already deviating is more likely to deviate further. It is included deliberately even at the risk of circularity because in many events, the deviation builds gradually over many bars before the label threshold is crossed, and the feature carries genuine forward information during that buildup.
- **s_net_flow_z_score_7d (mean |SHAP| = 0.4358).** The 7-day z-score of net mint/burn flow for coins with on-chain issuance data. Sustained net burning over a week (redemptions exceeding new mints) suggests that holders are exiting the coin; sustained net minting suggests new demand. The z-score normalization against coin-specific flow history allows comparability across coins with very different absolute flow volumes.
- **u_coin_type (mean |SHAP| = 0.2547).** Categorical feature identifying which of the seven coins the row belongs to. The model uses this to apply coin-specific baselines and to shift its prior on how likely each coin is to depeg in a given feature configuration.

UST, for example, receives a higher baseline prior than USDT under identical values of the continuous features.

The directional relationships between these features and the prediction are revealed by the beeswarm plot in Figure 6. Each dot is a single observation's SHAP value for a single feature; colour encodes whether that observation had a high (red) or low (blue) value of the feature.

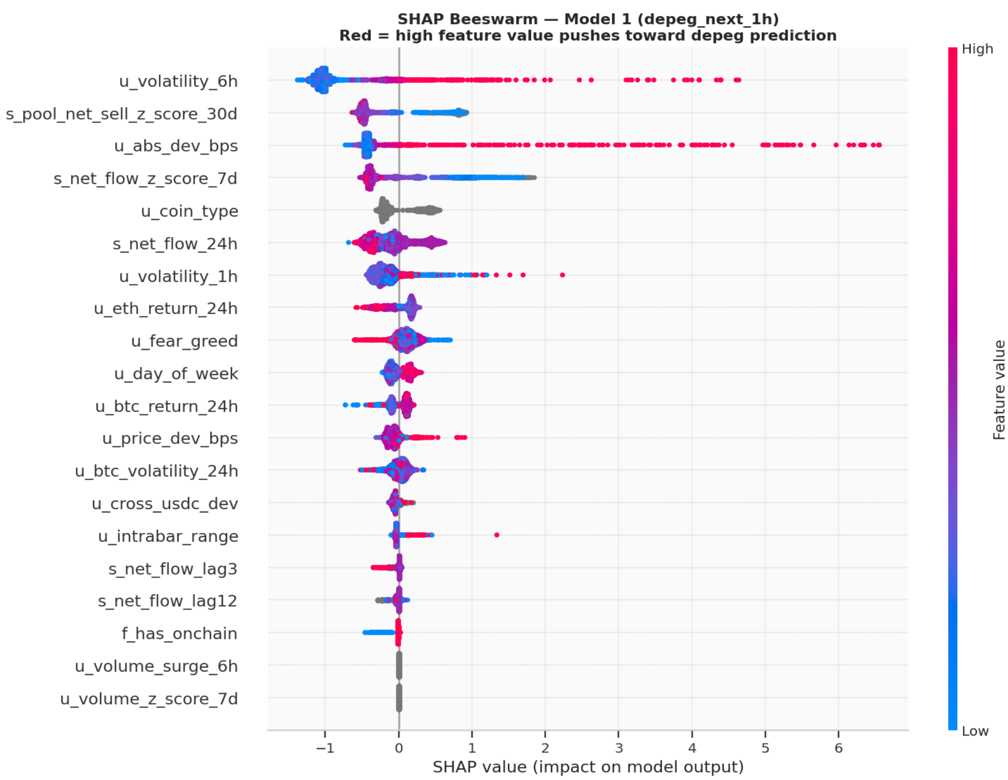

Figure 13. *SHAP beeswarm for the CatBoost model. Each dot is a single test-set observation. Horizontal position is the feature's SHAP value (contribution to the model's prediction). Colour encodes the feature value: red = high, blue = low. Clean colour separation (red on one side, blue on the other) indicates a monotonic feature-to-prediction relationship.*

Three directional patterns stand out. For u_volatility_6h, high values (red) produce strongly positive SHAP contributions — they push the prediction toward depeg — while low values (blue) produce negative contributions pushing toward stable. The separation is clean, indicating a consistent monotonic relationship. The same holds for u_abs_dev_bps where higher deviations mean higher depeg probability. For u_eth_return_24h, the relationship is inverted, high ETH returns (red) produce negative SHAP, pushing the prediction toward stable. A healthy broader crypto market suppresses stablecoin depeg risk, which is consistent with the interpretation that depeg events in this sample frequently coincide with general market distress. The s_net_flow_z_score_7d feature shows a more complex pattern where both strongly positive and strongly negative SHAP values exist with less clean colour separation, because the feature is bidirectional — extreme negative net flow and extreme positive net flow can both indicate stress, but through different mechanisms.

Predictive Signals

Moving from individual feature importance to the structure of the signal, three higher-level patterns emerge from the combined LOEO and SHAP analysis.

First, predictability is concentrated in events with gradual build-up. The Black Thursday DAI event produced a 23.2-hour lead time because its antecedents, cascading liquidations across DeFi driven by a 40% ETH price crash, unfolded over the preceding day in a way that continuously elevated volatility, intrabar range, and pool sell pressure features. By contrast, the May 2022 UST collapse had a zero-hour lead time despite a high AUC-PR of 0.5577, because the algorithmic depeg cascade happened within a single 5-minute bar once the threshold condition was breached. The model correctly identified UST as a high-risk coin, but it could not distinguish "high-risk UST about to break" from "high-risk UST just sitting above threshold" in the bars immediately preceding the crisis.

Second, the effective signal is primarily a price-dynamics signal augmented by a small number of on-chain features. Of the top five SHAP features, three are derived from price (volatility at two horizons, absolute deviation) and two from on-chain flow (pool sell pressure, net flow z-score). Macro features like u_fear_greed and u_btc_return_24h are secondary contributors. This has implications for feature engineering priorities, additional price-derived features (e.g., microstructure metrics at higher frequencies, order-book imbalance) would likely yield the largest performance gains, whereas further elaborations of macro sentiment are unlikely to move the needle much.

Third, cross-coin contagion is a real but secondary signal. The u_cross_usdc_dev feature, which measures how far USDC is from its peg, as a cross-coin stress indicator, appears among the top features with mean |SHAP| of 0.0511. This feature was specifically engineered to capture contagion risk, because stablecoin depegs often cluster in time (the USDC SVB event in March 2023 was preceded by USDT anomalies that same week). Its appearance in the top importance list validates the engineering decision, though its modest absolute contribution relative to the volatility and pool-flow features suggests that cross-asset contagion, while real, is a weaker signal than the coin's own internal dynamics.

Taken together, these patterns suggest a natural decomposition of depeg risk into three regimes. In quiescent regimes, all feature distributions look normal, and the model's output correctly sits near zero. In pre-crisis regimes, one or more of the top five features enter their tail — typically volatility first, then pool sell pressure, then absolute deviation — and the model output ramps up over minutes to hours. In acute crisis regimes, the price deviation feature itself becomes the dominant signal and the model locks onto coincident detection. An effective deployed system uses the same probability output differently across these regimes: as an early warning when features are entering their tails, as a confirmation when the deviation has taken over, and as a post-event monitoring tool once the initial break has happened.

Conclusion

This project set out to build an early warning system for stablecoin depeg events that is accurate, explainable, and operationally deployable. The final system meets each of these criteria to a usable standard, and the validation exercise, particularly the Leave-One-Event-Out cross-validation on five historical crises, establishes that the model's performance is grounded in genuine generalization rather than in fitting artifacts of the training distribution.

On accuracy, the CatBoost model achieves validation AUC-PR of 0.4965 against a random-baseline AUC-PR on the order of 0.0002. The F2-optimal threshold of 0.1273 produces precision of 0.349 on the test set, representing a 37× improvement over the base rate of 0.944%. At the event level, the model detected three of five LOEO crises early, with the earliest warning arriving 23.2 hours before the DAI Black Thursday collapse.

On explainability, the SHAP decomposition identifies five features that carry the majority of predictive weight. These are 6-hour price volatility, 30-day pool sell pressure, absolute price deviation from peg, 7-day net flow z-score, and coin type. Each of these is directly interpretable by a risk analyst familiar with stablecoin mechanics, and the directional relationships revealed by the beeswarm plots are consistent with prior domain knowledge. High volatility and sell pressure indicate depeg risk, while healthy broader market conditions suppress it. Every alert fired by the system can be accompanied by a short explanation naming the top three contributing features and their directional impact, which satisfies a practical regulatory explainability requirement.

On operational deployability, the system is lightweight in both training and inference. The CatBoost model occupies under 1 MB on disk, the Isolation Forest occupies 1.8 MB, and inference on a single 20-feature row is a matter of microseconds. The full pipeline from 5-minute bar ingestion to tier assignment can be run on a modest single-core server. The model artefacts are serialized to S3 and are ready for integration into any downstream alerting infrastructure.

The system's limitations are equally well understood. Three of five LOEO events produced coincident rather than early detections, because their precursors did not build gradually in the feature space. The small absolute number of labelled depeg events places an upper bound on model capacity that cannot be overcome by adding more training data from the same events. The pooled multi-coin architecture assumes feature-to-risk relationships that generalize across coin types, and this assumption breaks down partially for algorithmic stablecoins like UST. The threshold and tier cutoffs are defensible defaults but remain design choices that deployers will need to tune against their own operational constraints.

Several directions for future work follow directly from these limitations. First, incorporating higher-frequency market microstructure features — order book depth, bid-ask spread, trade imbalance at sub-second resolution — would likely improve early-warning performance on events that currently produce zero-lead detections. Second, augmenting the training set with semi-synthetic depeg events, generated by perturbing historical price paths in controlled ways, could partially address the small-positive-class limitation. Third, a meta-learning architecture that trains per-coin-type models and ensembles them via a stacking layer could better accommodate the heterogeneity between fiat-backed, crypto-collateralized, and algorithmic stablecoins. Fourth, extending LOEO to include synthetic events or events in related asset classes (for example, currency pegs, money market fund breaks) would test and potentially improve the model's generalization beyond the stablecoin-specific training distribution.

Beyond the specific findings, the broader contribution of this work is methodological. It demonstrates that a carefully engineered, leakage-controlled, explainability-first modelling pipeline can produce actionable early warnings for a rare-event financial risk problem using publicly available data and open-source tooling. Stablecoin depeg prediction is a narrow problem, but the pipeline developed here is a general-purpose template for any domain where the events that matter are rare, costly, and only partially predictable from their precursors.

The question posed at the start of this project was whether a model could see a depeg coming. The answer, with appropriate caveats, is yes — for the class of events whose precursors build gradually in the visible feature space. For the class of events that arrive as shocks, the model's value is in confirmation and post-event monitoring rather than in early warning. Either way, the system represents a meaningful improvement over the alternative of no model at all. In a market where a missed warning can cost tens of billions of dollars and a false alarm costs a single unnecessary hedge, that improvement is worth operationalizing.

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